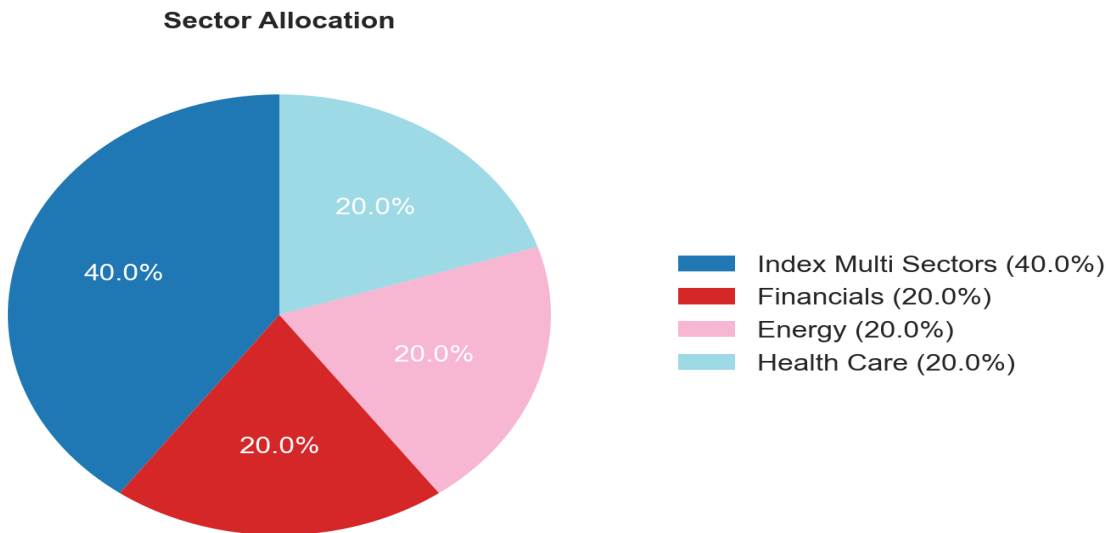


Portfolio rotation report

Date of extraction	2025-12-07 21:22
Data window	2019-09-03 → 2020-06-29
Frequency	Daily — 208 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+4.0% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	QQQ, SPY, XLE, XLF, XLV
Indicators	Short term: 20 days volatility, 41 days return, 62 days momentum Long term: 50 days volatility, 82 days return, 186 days momentum

1. Portfolio Composition

- XLF (Financial Select Sector SPDR Fund — Financials)
- XLE (Energy Select Sector SPDR Fund — Energy)
- XLV (Health Care Select Sector SPDR Fund — Health Care)
- SPY (SPDR S&P; 500 ETF — Index Multi Sectors)
- QQQ (Invesco QQQ Trust — Index Multi Sectors)



Macro Profile Insight

Macro coverage

- Partial coverage: several macro themes remain unrepresented.
- Broad balance across market capitalizations: diversification achieved across the size spectrum.

Structural properties

- Concentration risk: portfolio heavily exposed to etf instruments.
- Liquidity stance: pockets of thin/illiquid names; robust median dollar volume.
- Market-neutral beta exposure: close alignment with index dynamics.
- Balanced volatility structure: risk evenly distributed across holdings.

Additional comments

- Significant sector gaps: broad portions of the economy remain uncovered.

2. Sector Performance Visualization



Top Positive Correlations

100.0% of portfolio

Pair	Correlation
QQQ / SPY	0.96
SPY / XLF	0.95
SPY / XLV	0.93
QQQ / XLV	0.91
XLE / XLF	0.88

Overall, average cross-sector correlation was 0.86, ranging from 0.71 to 0.96. This suggests very high correlation — sectors behave almost as a single macro block. Correlation dispersion is narrow, suggesting similar risk dynamics across sectors. Strongest co-movement between QQQ and SPY (0.96); weakest between XLE and XLV (0.71). Market-driven regime: macro factors dominate individual sector dynamics.

3. Key Performance Indicators

Ticker	R (%)	V (%)	RAR	M (%)	Signal	Regime	Liquidity	Commentary
QQQ	0.26	1.66	0.156	0.45	neutral	subdued	liquid	mixed short-term dynamics, signal unclear or in transition, low volatility — calm market tone, with ample liquidity.
SPY	0.13	1.91	0.066	0.30	neutral	subdued	liquid	mixed short-term dynamics, signal unclear or in transition, low volatility — calm market tone, with ample liquidity.
XLE	-0.03	4.00	-0.008	0.47	neutral	subdued	liquid	recovering momentum despite recent losses, signal unclear or in transition, low volatility — calm market tone, with ample liquidity.
XLF	0.02	3.11	0.006	0.17	neutral	subdued	thin	mixed short-term dynamics, signal unclear or in transition, low volatility — calm market tone, under limited liquidity.
XLV	-0.02	1.71	-0.013	0.19	neutral	normal	thin	recovering momentum despite recent losses, signal unclear or in transition, normal volatility regime, under limited liquidity.

Top by Average Return

Ticker	AvgReturn(%)
QQQ	0.26
SPY	0.13
XLF	0.02
XLV	-0.02
XLE	-0.03

Top by Volatility

Ticker	Volatility(%)
XLE	4.0
XLF	3.11
SPY	1.91
XLV	1.71
QQQ	1.66

Top by Momentum

Ticker	Momentum(%)
XLE	0.47
QQQ	0.45
SPY	0.3
XLV	0.19
XLF	0.17

3.b Rotation & Stress Metrics

Ticker	ΔReturn (pp)	Momentum Ratio	Signal	Volatility Ratio	Beta	ADV10 USD	Interpretation
QQQ	0.13	2.56	0.127	1.03	1.18	53.05 M	stable performance, momentum surging, volatility normalizing, within defensive profile, under thin liquidity regime.
SPY	0.23	3.00	0.075	1.14	1.00	72.91 M	stable performance, momentum surging, volatility normalizing, within defensive profile, under thin liquidity regime.
XLE	0.36	-2.98	-0.118	1.12	0.80	28.61 M	stable performance, momentum fading, volatility normalizing, within defensive profile, under thin liquidity regime.
XLF	0.31	-1.70	-0.063	1.14	0.84	36.84 M	stable performance, momentum fading, volatility normalizing, within defensive profile, under thin liquidity regime.
XLV	-0.12	1.90	0.066	1.18	0.48	15.05 M	stable performance, momentum surging, volatility normalizing, within defensive profile, under thin liquidity regime.

■ Light-impact observations

For **QQQ, SPY, XLE, XLF**: *Volatility compression* — subdued volatility; a breakout setup may be forming. (avg ann. return 0.1%, momentum 0.3%, vol 2.7%)

For **XLV**: *Range / noise* — no clear direction; rotating momentum and mid-range volatility. (avg ann. return -0.0%, momentum 0.2%, vol 1.7%)